

2(a)(i)	$P = Fv$	C1
	$= 18 \times 1.4$	A1
	$= 25 \text{ W}$	
2(a)(ii)	$a = F / m$	C1
	$a = 18 / 72 = 0.25 \text{ (ms}^{-2}\text{)}$	C1
	$t = 1.4 / 0.25$	
	$= 5.6 \text{ s}$	A1
2(b)(i)	$a = (54 - 18) / 72 \text{ or } 36 / 72 (= 0.50 \text{ ms}^{-2})$	C1
	$v^2 = 2 \times 0.50 \times 9.5$	C1
	$v = 3.1 \text{ m s}^{-1}$	A1
2(b)(ii)	$W = 54 \times 9.5$	A1
	$= 510 \text{ J}$	
2(b)(iii)	curved line from the origin	M1
	gradient of line increases	A1
2(c)	(force due to) air resistance increases/changes/not constant or air resistance increases with speed	B1